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Title:

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Date: 2024

Type: Article de revue / Article

Référence: Pierre, J., Verville, M., Chenier, G., Farahani, R. D., Piccirelli, N., Lévesque, M., & Therriault, D. (2024). Non-planar material-extrusion additive manufacturing of multifunctional sandwich structures using carbon-reinforced polyetheretherketone (PEEK). Additive Manufacturing, 84, 104124 (13 pages).
Citation: <https://doi.org/10.1016/j.addma.2024.104124>

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Révisé par les pairs / Refereed

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Document issued by the official publisher

Titre de la revue: Additive Manufacturing (vol. 84)
Journal Title:

Maison d'édition: Elsevier
Publisher:

URL officiel: <https://doi.org/10.1016/j.addma.2024.104124>
Official URL:

Mention légale:
Legal notice:

Non-planar material-extrusion additive manufacturing of multifunctional sandwich structures using carbon-reinforced polyetheretherketone (PEEK)

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Highlights

- A 6-DOF robot prints carbon-reinforced PEEK non-planar structures in a heating enclosure.
- Non-planar curved beams are 28% stiffer than planar curved beams in bending.
- A non-planar sandwich panel, with 2 types of microscallops and acoustic cells, was printed.

Abstract

Additive Manufacturing (AM), and especially Material-Extrusion Additive Manufacturing (MEAM), can provide light structures with high mechanical performances for aerospace applications using fiber reinforced high-temperature resistant thermoplastics (HTRT). Non-planar AM allows to manufacture curved parts with a better approximation of their curvature than cartesian AM. In this paper, we additively manufactured a proof of concept of multifunctional non-planar sandwich panel for aircraft casing using a 6-degree of freedom (6-DOF) robotic platform in a custom heating enclosure, using 30 wt.% carbon-reinforced polyetheretherketone (PEEK). Curved beams manufactured using planar and non-planar

deposition techniques were manufactured using a custom heating enclosure and reinforced HTRT. The measured surface deviation of the non-planar curved beam (NPCB) was 31% lower than the planar curved beam (PCB). Due to the non-planar deposition that led to a better alignment of the layers along the specimen, the effective stiffness measured during 3-point bending tests was also increased by ~28% compared to PCB. Additively manufactured non-planar sandwich panels containing an acoustic core and two types of microscallops (30% infill microscallops and 78% infill walled microscallops) were characterized through optical microscopy and 3-dimension (3D) scanner measurements. The 6-DOF robotic platform, combined to the heating enclosure, allowed to print a portion of a non-planar carbon fiber reinforced PEEK sandwich panel to present the versatility of the structures that could be printed with this setup. The use of non-planar reinforced HTRT structures could find applications in the aerospace field for aircraft noise reduction structures, but also in the biomedical domain for the manufacturing of biocompatible PEEK scaffolds.

Abbreviations

3D, three dimensional

6-DOF, six degrees of freedom

ABS, acrylonitrile butadiene styrene

AM, additive manufacturing

C2M, CAD to Mesh

CAD, computer aided design

CB, curved beam

CI₉₅, 95% confidence interval

CFR-PEEK, carbon fiber reinforced polyetheretherketone

CLMS, Curved Layer Mesh Slicing

CNC, computerized numerical control

CT-scan, Computerized Tomography-scan

FFF, fused filament fabrication

HTRT, high temperature resistant thermoplastics

MEAM, material-extrusion additive manufacturing

NPCB, non-planar printed curved beam

NURBS, non-uniform rational B-spline

PLA, polylactic acid

PCB, planar printed curved beam

PEEK, polyetheretherketone

PEI, polyetherimide

STL, standard triangle language

TPP, teach pendant programing

Keywords

Additive manufacturing, Fused Filament Fabrication, Non-planar printing, High-temperature resistant thermoplastics, Carbon fiber-reinforced composites

1. Introduction

Additive Manufacturing (AM), also known as three-dimensional (3D) printing, is a family of technologies that allows the layer-by-layer fabrication of parts. Recent advances in the AM of high temperature resistant thermoplastics (HTRT), such as PEEK [1] or polyetherimide (PEI) [2], not only allow rapid prototyping, but also the creation of components featuring multifunctionality and complex geometries that usually cannot be achieved using conventional manufacturing methods (e.g., injection molding).

Material-Extrusion Additive Manufacturing (MEAM), and especially Fused Filament Fabrication (FFF), is widely used and compatible with a large variety of thermoplastic materials

and reinforced thermoplastic composites, such as polylactic acid (PLA), nylon, or even HTRT [3]. Reinforcing thermoplastics with fillers [4] [5] [6] and especially short carbon fibers [7] [8] is a mean to significantly improve their mechanical performance. Common commercial FFF printers extrude melted thermoplastic filament, thereby creating a vertical stacking of flat layers, and are therefore referred to as cartesian 3D printers. They typically mobilize up to two axes simultaneously, using the third one to change layers. However, using FFF cartesian and the planar printing approach can create geometrical inaccuracies, referred to as “staircase effect”, in which the curvature is approximated by the stacking of discrete layers [9] [10]. A method called non-planar printing, consisting of depositing layers along the part curvature, not only improves the geometrical accuracy, but also increases the mechanical properties of the printed parts [11]. Using acrylonitrile butadiene styrene (ABS) and a cartesian printer, Singamneni et al. compared the mechanical properties of planar and non-planar printed curved beams, and showed that the beams printed in a non-planar way supported a maximum load 40% higher than the beams produced by planar printing [12]. Non-planar printing can be performed using cartesian 3D printers by customizing the G-code (i.e., the language containing motion and extrusion instructions for the printer) to simultaneously use the three motion axes of the printers [13]. However, the angle of deposition is rather limited due to the typical shape of the printing head. If the printed layer’s curvature is too pronounced, the nozzle can damage the underlying part by inducing pronounced local gouging.

Recent efforts have focused on the utilization of 6-degree of freedom (6-DOF) robotic systems as a printing platform to enable printing over non-planar surfaces [14] without local gouging. Several robotic systems have already been combined to FFF print heads to additively manufacture parts and test specimens, using polymers such as PLA [15] [16] or ABS [17]. A 6-DOF robotic platform also enables adjusting the angle of deposition and therefore reducing the need of support structures [18] [19], unlike cartesian 3D printers [13] [20]. Depending on

its size, the 6-DOF robotic platform can be used to manufacture large FFF parts up to a meter long [21]. They can also be compatible with fast extrusion systems, such as pellet extruders [22]. The 6-DOF robotic platform, however, requires additional programming steps. In addition to printing instructions, motion instructions for each of the robotic joints need to be computed [23] [24], as well as a toolpath avoiding collisions between the printhead and the printed part [25].

The printing of parts using HTRTs not only requires a high temperature extrusion system (e.g., a high temperature extrusion nozzle) and a heating bed, but also a heated environment [26] [27]. The elevated temperature provided by a heating enclosure reduces the printed part residual deformation (e.g., warping), increases the mechanical properties of the part as it improves the weak interfilament interface [28], and reduces the interlayer porosity [29] [30]. Several cartesian 3D printers (e.g., AON M2+ from AON3D, Gearbox HT2 from 3DXtech) feature an integrated heating system, allowing the users to manufacture HTRT parts [31]. A few examples of 6-DOF robotic platforms printing HTRT exist, but they are designed to manufacture parts with very large nozzles [32] [33]. For example, Atzler et al. manufactured a non-planar sandwich panel using granulated carbon-fiber reinforced PEEK (CFR-PEEK) for the honeycomb core and laminate sheets for the facesheets [34], but used a large nozzle, which prevents the fabrication of fine details and the extrusion of thin walls ($<1\text{mm}$).

Combining non-planar FFF with HTRT materials could enable the fabrication of custom parts for aerospace applications [35]. For example, traditionally manufactured sandwich panels with honeycomb core structures [36] are located on the fan case around some aircraft engines [37]. They have a noise reduction purpose [38], but being able to manufacture non-planar sandwich structures with a custom acoustic core design [39] and porous microscavolds (e.g., periodic staking of filaments) [40] could provide enhanced acoustic properties. The use of reinforced HTRT could provide loadbearing functionality to the structure as well as thermal resistance

[41]. There is therefore a growing interest to develop FFF methodologies to print reinforced HTRT parts whose geometry yield noise reduction and can withstand mechanical loads as well as heat exchanges. To the best of the authors' knowledge, no non-planar complex multifunctional sandwich structure has been fully printed using CFR-PEEK and a 6-DOF robotic platform.

In this paper, we have successfully manufactured by FFF HTRT non-planar curved beams and a non-planar multifunctional sandwich structure using a customized 6-DOF infrastructure for FFF. The multifunctionality of the sandwich panel is demonstrated by the realization of different acoustic geometries (i.e., acoustic core and micro scaffold) and the use of reinforced HTRT and non-planar FFF for enhanced mechanical loadbearing capacities.

2. Materials and methods

2.1. AM platform with 6-DOF robot

The AM platform consisted of a 6-DOF industrial robot (FANUC M-20iB/25, FANUC Canada Ltd., Canada) on which an FFF extrusion system designed for HTRT prints was installed. Figure 1a presents a schematic of the AM platform, composed of the 6-DOF robot, a 3D printing bed, and a custom heating enclosure with an infrared heater. The printing bed was an aluminum plate (AON3D, Canada) (450 mm $\times$ 450 mm) with a silicon rubber pad heater (SainSmart, USA) fixed underneath. The printing bed was covered with polyimide adhesive film (Kapton, Dupont, USA). The heating enclosure was built using T-slotted aluminum extruded shapes on which insulating plates of polyisocyanurate R-6 (Atlas E148, Atlas, Canada) were mounted, covering all the sides and the bottom of the structure. A reflective mylar film (Vivosun, USA) was applied on the interior of the sidewalls. The dimensions of the enclosure were 90 cm $\times$ 80 cm $\times$ 50 cm. A 3.75 kW ceramic infrared heater (CRP20074, Omega, Canada) was mounted to one of the sidewalls inside the enclosure. The bottom of the heater was fixed at a height of 3

cm above the printing bed. The heater system used a feedback loop in which a K-type thermocouple, attached inside the enclosure, allowed to adjust the temperature of the heater until the desired internal target temperature was reached. The bed temperature was controlled separately.

Figure 1b shows the FFF printing head system consisting of a water-cooled high temperature FFF extruder and hotend (Dyze Design Pro, Dyze Design, Canada). This extrusion system was mounted on a computerized numerical control (CNC) machined PEI support plate attached to the wrist of the 6-DOF robot.

Figure 1c shows the silicon tarp (Rubber-Cal, USA) covering the top-opening of the enclosure to allow the printhead to move within the printing area. The different sections of the blanket were attached with straps (Velcro, United Kingdom). This system allowed us to easily reach the inside of the enclosure and remove the printed parts. A high temperature resistant bracket was designed and additively manufactured using PEI9085 (3DXtech, USA) to attach the silicon blanket to the 6-DOF robot through the PEI holder. Six retractable tension cables were attached between the AM bracket and the top of the enclosure's sidewalls to prevent the blanket from falling on the part being printed or from moving too close to the heating bed or heater. Figure 1d shows a picture of the experimental setup with the tarp opened. The silicon tarp was divided into four sections, each fixed to a side of the enclosure with press buttons. A small window was installed in one of the side walls to monitor the printing during the experiment with a webcam. A metal tube attached to the 6-DOF robot was used as a spool holder.

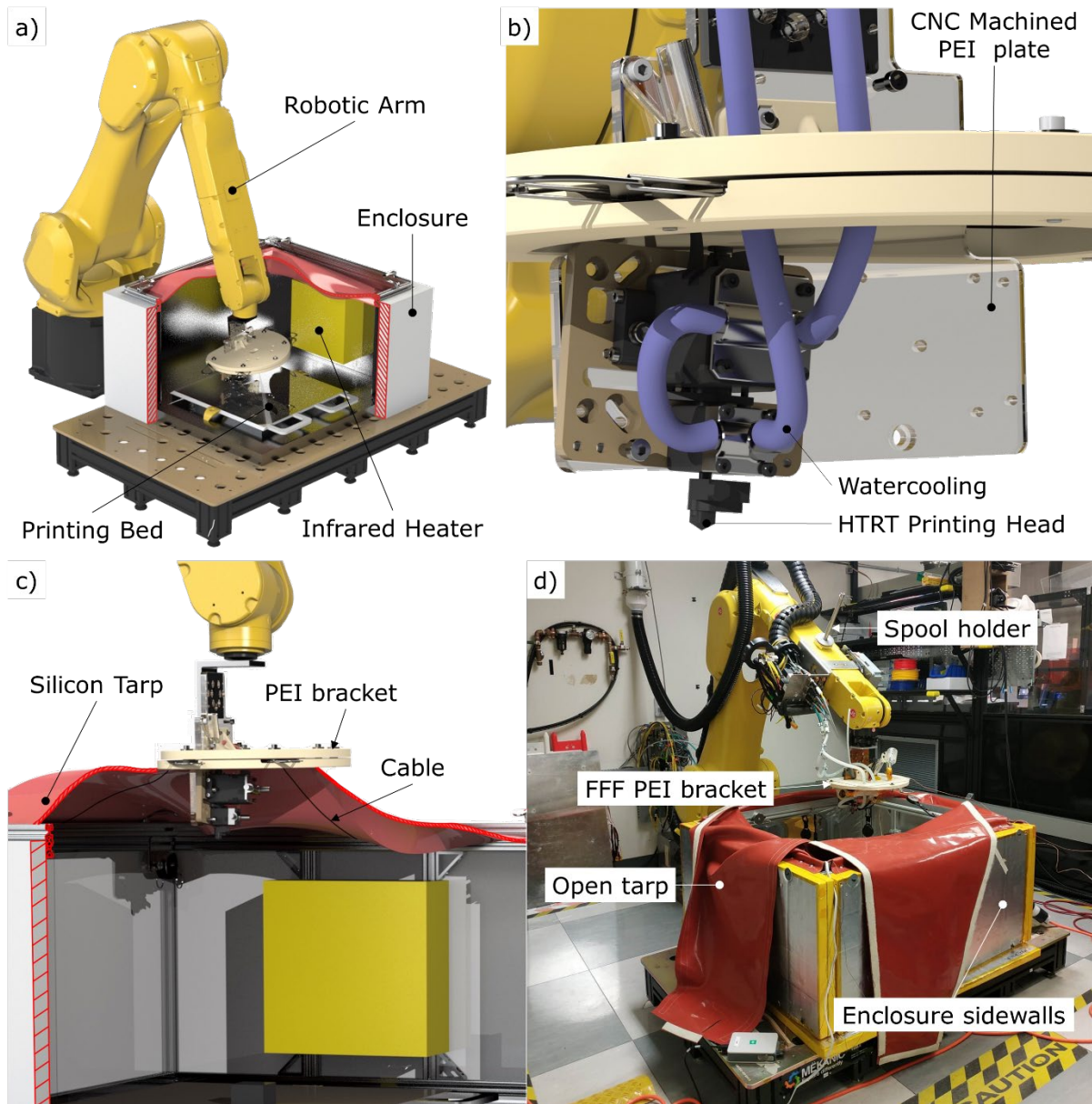


Figure 1: AM platform with 6-DOF robot for HTRT FFF. a) Schematic of the 6-DOF robot and the heating enclosure (walls and blanket have been cut to show the inside of the chamber). b) Schematic of the FFF printing device attached to the robot wrist. c) Schematic of the inside of the chamber, with a holder to fix the silicon tarp to the enclosure and the robot, and cables to prevent the tarp from touching the hot surfaces. d) Picture of the AM platform with the silicon tarp opened.

2.2. Additive manufacturing process

Using a custom 6-DOF robotic platform to manufacture planar and non-planar specimens required additional steps, when compared to cartesian FFF printing. Figure 2 shows the whole process, divided in three main phases: shape generation, slicing, and printing. Three types of

specimens were manufactured during this work: a curved beam (CB) printed with a planar method (PCB), a non-planar curved beam (NPCB), and non-planar sandwich panels.

Shape generation

Curved beam specimens inspired by [12] were designed using a computer aided design (CAD) software (Catia V5-6R2017). Support structures were also manually designed for the NPCB.

An aircraft casing with a double curvature surface along the x and y axes, colored in dark grey in the schematic, was chosen as the substrate for the manufacturing of the sandwich panels. The casing was scanned with a 3D scanner (GOM Atos Core 200, GOM Metrology, Germany) and was reconstructed as a non-uniform rational B-spline (NURBS) surface, shown in red. Supports for the sandwich panels were then designed using a CAD software (CATIA V5-6R2017) to replicate the scanned surface of the casing. Sandwich panels were designed using a CAD software (Catia V5-6R2017) with the NURBS surface guiding the geometry of the bottom of the panels. All files were saved in standard triangle language, or .STL, files.

Slicing

The curved beams were printed using two distinct methods: planar FFF and non-planar FFF. PCB were directly sliced with a commercial software (Simplify 3D, version 4.1.2), which added support structures underneath and converted the .STL file into a G-code file (containing motion and printing instructions). Support structures for non-planar specimens (NPCB and sandwich panels) were also sliced with Simplify3D. NPCB and the sandwich panels required a custom slicing method called Curved Layer Mesh Slicing (CLMS). The print head was oriented to print the filament on top of the previous layer while following the curvature of the layer, and to avoid gouging [13]. Slicing surfaces were created by interpolating between the top and bottom surfaces of the part. Repeating the interpolation operation several times created all the slicing surfaces required for every layer of the part. For the NPCB, since the thickness of the beam was constant, all the layers had the same thickness and followed the curvature of the beam. For the

sandwich panel, the bottom surface had a double curvature along the x and y axes. However, the upper surface of the sandwich panel needed to be perfectly cylindrical along the x axis to fit the aircraft requirements. To keep a constant number of layers despite the variation of thickness through the panel, the slicer calculated a variable layer thickness within the same layer, as shown in the schematic in Figure 2.

With the slicing surfaces established, mesh-mesh intersection operations computed the intersection lines between the slicing surfaces and the STL mesh model. The resulting intersection lines were collected to generate the outline toolpaths. Non-planar infill was created on the inside of the outlines, as shown by the red arrows on the schematic.

Once a G-code file was generated for the specimens, it was imported to the robotic simulation software (RoboDK, ver. 5.4.1) to solve the inverse kinematic problem for each coordinate of the toolpath. Although the robot could rotate along 6 degrees of freedom, a compromise had to be found to avoid collisions between the 6-DOF robot, the enclosure, and the twisting of the cables around the arm.

The information was then transmitted to the 6-DOF robot motion controller (R30iB Plus, FANUC, Canada) via a teach pendant programming (TPP) file (i.e., proprietary file format for FANUC).

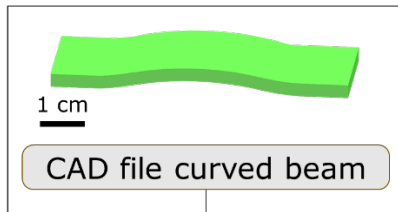
Manufacturing

The robot motion controller sent the motion instructions to the robot and the printing instructions to the FFF printhead controller (Mëkanic, Canada), which delivered the extrusion instructions to the printhead.

Due to their traditional slicing, PCB were manufactured at the same time as their supports. For NPCB and non-planar sandwich panels, supports were fully printed first, and then, non-planar specimens were printed on top of them.

Shape Generation

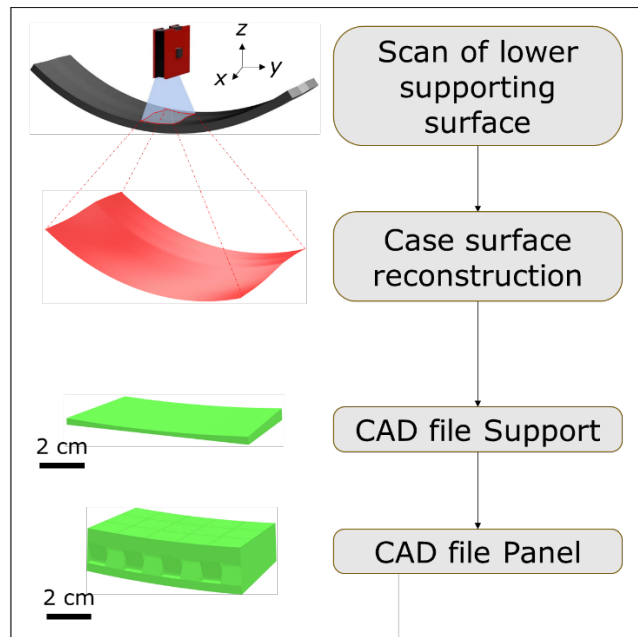
Curved Beams



CAD file curved beam

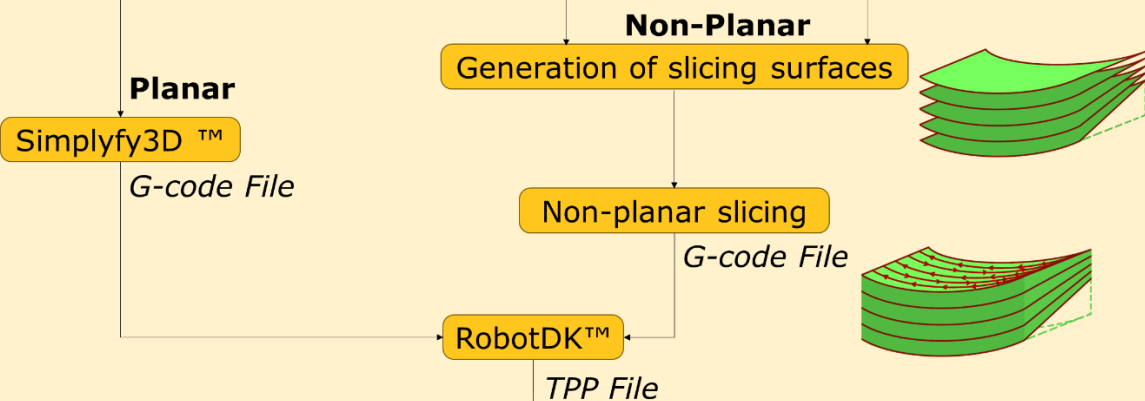
STL file

Sandwich Panel



STL file

Slicing



Printing

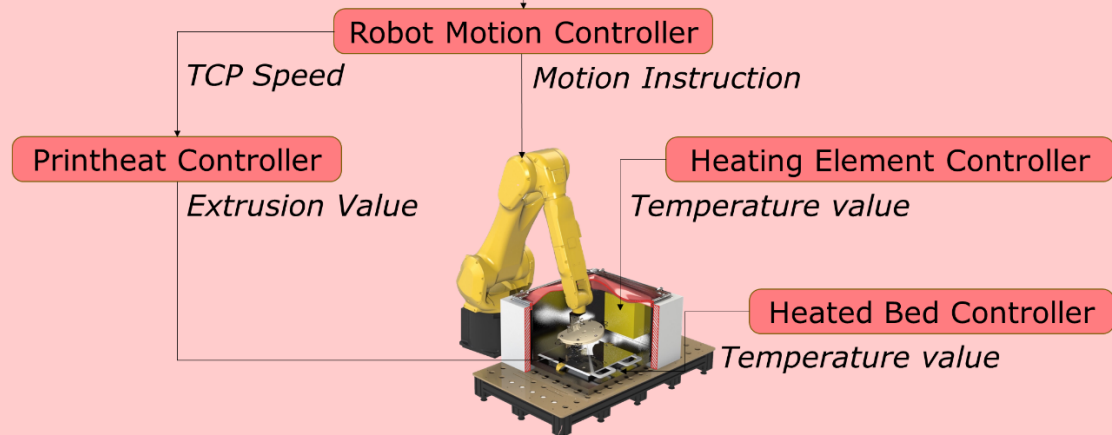


Figure 2: Process of FFF printing with the 6-DOF robot divided in three main steps: shape generation, slicing, and manufacturing. Two types of geometries were printed: curved beams and a sandwich structure. Two slicing paths were used: planar and non-planar (CLMS). The printing platform is the same for all specimens.

2.3. FFF printing parameters

A 1.75 mm diameter spooled filament of PEEK reinforced with 30 wt.% short carbon fibers (TECAPEEK ® CF30 black, ENSINGER, Germany) was used to print the specimens. The printing nozzle had an inner diameter of 0.4 mm. Before each print, nano polymer adhesive (Vision miner, USA) was spread on the printing surface to prevent decohesion with the substrate. Table 1 regroups the main printing parameters of the curved beams and the sandwich structures.

Table 1 Values of the printing parameters used for the creation of the HTRT structures.

Type of specimen	Beams	Sandwich structures
Printhead temperature (°C)	440	440
Bed temperature (°C)	140	140
Extrusion multiplier	0.9	0.9
Extrusion width (mm)	0.38	0.38
Layer thickness (mm)	0.2	0.09-0.25
Printing speed (mm/s)	15	15
Shell	n/a	1

The measurements of the bed and internal enclosure temperatures are found in Supplementary Figure S1 and Supplementary Table S2, showing the increase of the temperature in the chamber up to ~120°C and its stabilization after ~45 minutes of heating.

Five specimens of PCB and five specimens of NPCB were manufactured, as well as two curved sandwich panels, and a portion of a curved sandwich panel displaying its internal structure. The beams had an infill percentage of 100% and a layer orientation at 0-90°.

Figure 3a shows a schematic of the curved beams and their support structures. The beams were printed with the convex surface facing up since printing with the concave surface facing up would have induced multiple collisions between the 6-DOF robot and the enclosure. Figure 3b shows the orientation of the layers along the x and y axes and an illustration of a PCB and a

NPCB (only a few layers have been represented for the sake of clarity). The orientations were 90° along the y axis for the blue layers and 0° along the x axis for the yellow layers. The layers of the PCB were printed planarly along the xy plane, unlike the NPCB, whose layers followed the curvature of the beam.

Figure 3c shows a schematic of the sandwich panel in its original flat configuration, which is made of different zones stacked together. Starting from the bottom, the lower facesheet of the panel, shown in red, consists of a 1 mm thick 100% infill with layers following a 0- 90° orientation pattern. Above that, the core, shown in purple, is composed of 16 mm-high trapezoidal hollow thin-walled structures having shown acoustical performance [39] (more details about the structure can be found in Supplementary Figure S3), with a wall thickness of 0.4 mm (which is approximately the nozzle's diameter). Then, a first type of microsccaffold is added on top (microsccaffold 30% in light blue), with a targeted infill percentage of 30% and a thickness of 8 mm. A 30% infill means that in a given layer, extruded filaments are spaced evenly to cover 30% of the area. Finally, a walled microsccaffold (walled microsccaffold 74%) covers the top of the panel. The walls are represented in orange and contain a 74% infill scaffold (in dark blue) enclosed in $24\text{ mm} \times 24\text{ mm}$ squares.

The flat panel configuration needed to be adapted to fit the casing surface. The casing surface presented a double curvature, along x and y axes, as seen in the side views of Figure 3d. A non-planar adaptive set of layers (in yellow) was added to the bottom of the panel structure to compensate for the y curvature (highlighted with a black dotted line on the casing surface) so that the upper surface of the panel would have a single curvature. This yellow zone was referred to as “adaptive” because in the same layer, the thickness gradually varied between 0.09 mm and 0.25 mm by adjusting the material extrusion rate using our custom slicer. It allowed to remove any staircase effect induced in a part that would not have a constant thickness. All layers (except in the purple zone) were oriented at 0- 90° . The total thickness of the adaptive layer set

varied along the x axis, between 2.0 mm and 5.8 mm, enabling its upper surface to be only cylindrical along the x axis. The geometry of the flat panel was then curved in the CAD software to fit the upper surface of the adaptive layer. Due to the adaptive layer set, the curved panel was slightly thicker than the flat one (29 mm for the flat panel, compared to between 31.0 mm and 34.8 mm depending on the position along the x axis). The final dimensions of the curved panel were 125 mm in length with a curvature radius of 967 mm $\times$ 36 mm of width $\times$ between 31.0 mm and 34.8 mm of thickness.

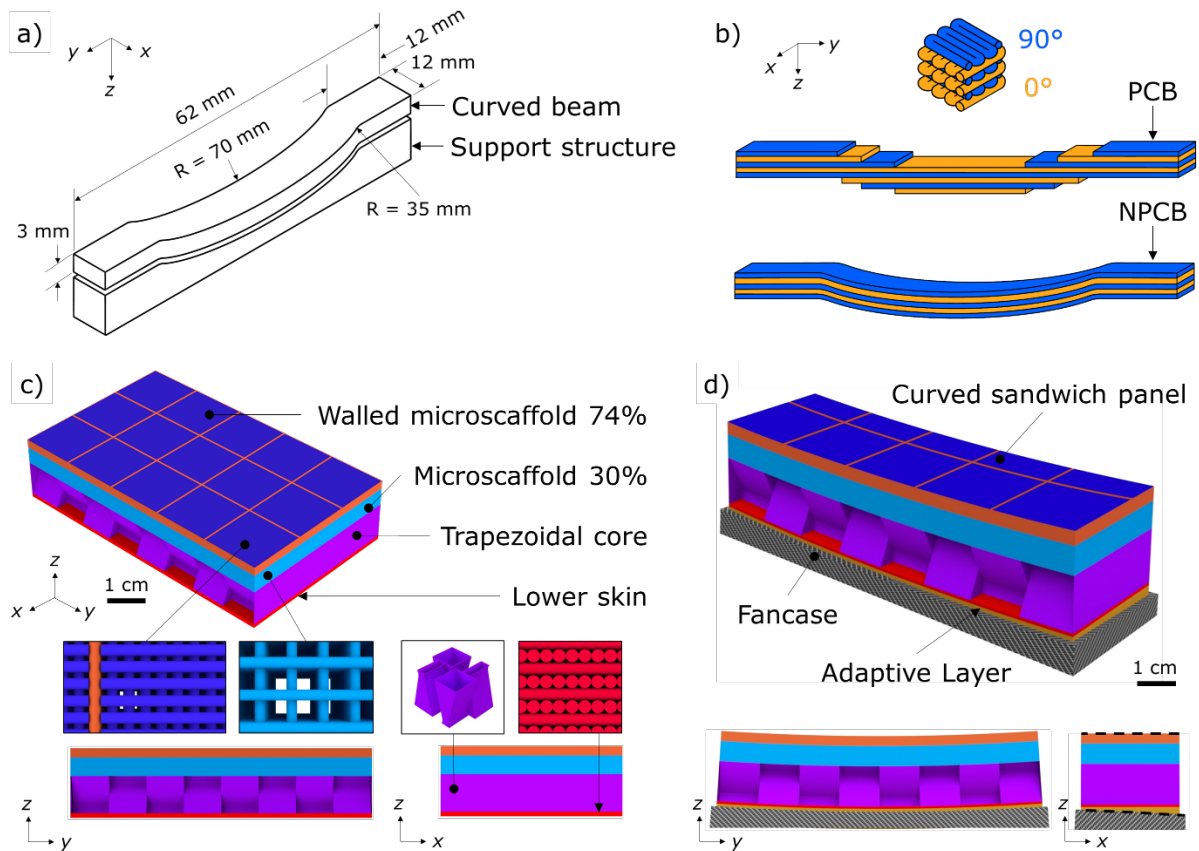


Figure 3 Schematics of the geometry of the specimens a) Curved beam and its supports. b) Illustration of the orientation of the printed layers in the beams: [0/90], a PCB and a NPCB on which some layers and their orientation have been represented. c) Original flat sandwich panel from different viewing perspectives. It is composed of a 74% infill walled microsccaffold, a 30% infill microsccaffold, a core of trapezoidal hollow shapes and a 100% infill lower facesheet. d) Sandwich panel once curved from different angles: an adaptive layer must be added at the bottom of the panel to compensate for the casing curvature visible in the xz plane (black dotted line).

2.4. Characterization

Optical microscope

The top surface of the specimens was observed using a stereo microscope (Olympus SZX12, Olympus Corporation, Japan). Measurements were made using an image processing software (ImageJ, version 1.52a).

3D Scanner

The specimens surfaces (curved beams and panels) were scanned using a 3D scanner (GOM Atos Core 200, GOM Metrology, Germany), reconstructed as a mesh and saved as an .STL file. The deviation of the printed specimens was assessed by measuring the signed (either positive or negative) distance between the CAD of the specimen and its reconstructed mesh (CAD to Mesh or C2M) with the software CloudCompare (version 2.12.4).

Scanning electron microscope (SEM)

Curved beams specimens were observed with a tabletop SEM (TM3030 Plus, Hitachi, Canada). Before the observation, the specimens were covered with a 20 nm chromium plasma coating to improve the quality of the picture.

Computerized Tomography-scan (CT-scan)

Tested curved beams were observed with a CT-scan (Xradia 520 Versa, Zeiss, Canada) at an acceleration voltage of 50 kV. Two different pixel sizes of 43 microns and 14.11 microns were used. A visualization software (Dragonfly, Object Research Systems ORS Inc, Canada) was used to generate the CT-scan pictures shown in this paper.

2.5. Mechanical testing

Three-point bending tests were conducted using an MTS Insight (Eden Prairie, USA) testing machine with a 1 kN load cell. The deflection speed of the cross head was 2 mm/min. The effective stiffness was computed through a linear regression of the load/displacement curves. The beams were placed upside down when compared to their printing orientation [12].

2.6. Microscaffold characterization

The top facesheet of the sandwich panel was composed of two types of microscaffolds. The printed microscaffolds were observed using an Olympus SZX12 stereomicroscope. The consistency of the areas A of each squared void, when looking at the scaffold from above, was measured using the image processing software ImageJ. A method for measuring the printability was adapted from Brzeski et al [42]. The printability S in the xy plane was defined as:

$$S = \frac{1}{n} \sum_{i=1}^n 1 - \frac{|A_n - \bar{A}|}{\bar{A}}, \quad (1)$$

with n the number of measured areas, A_n the measured area, and $\bar{A}$ the average measured area.

Figure 4 shows three examples of scaffolds with different values of S . A 30% microscaffold with an S of 1 indicates that its shape retention is quasi-perfect. From a top view, all voids created by the stacking of filaments have the same area, and no defects are visible in the xy plane. In Figure 4b, even if the pattern is quasi-perfect, one square is filled, as it can happen with dust or a blob of material, which leads to a decrease of the coefficient. Changing the spacing between the filaments also affect S since the areas of the voids are no longer identical, like the example in Figure 4c. The threshold of acceptable printability was arbitrarily set to 0.95.

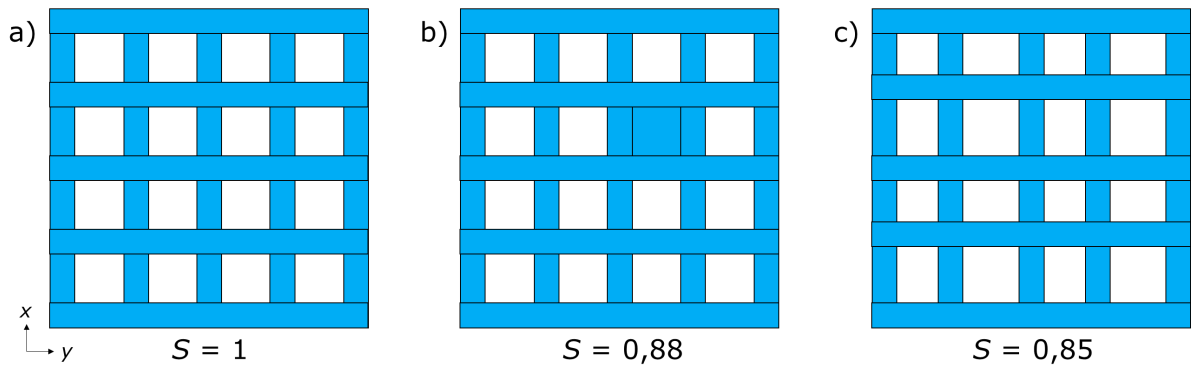


Figure 4 Example of different top view of microscaffolds with 30% infill featuring manufacturing defects with printability index S . a) $S = 1$: scaffold is perfectly repeated. b) $S = 0.88$: scaffold is perfectly repeated, but a square has been filled with impurities, making the coefficient decrease. c) $S = 0.85$: varying the spacing between the filaments decreases S further.

3. Results and discussion

3.1. Characterization of the planar and non-planar curved beams

Figure 5a shows top and side view images of PCB (left) and NPCB (right). From the top view, the stacking of layers in the 0° and 90° orientations is clearly visible on the PCB, whereas only one layer in the 90° direction is visible on the NPCB. From the side view, the 0° curved layers of NPCB cover the length of the whole specimen, unlike the 0° layers of PCB. Figure 5b shows the upper surface of the curved beam CAD, a scanned surface of a PCB, and a scanned surface of a NPCB. The scanned surfaces have been colored according to the C2M signed distance. The PCB scan colors are contrasted, showing a variation of the C2M signed distance, and the staircase pattern is visible. The strong red color visible on the side of the steps could result from an accumulation of material due to over extrusion. The NPCB is less contrasted, however. Figure 5c shows a graph comparing the C2M signed distances of the scanned surfaces of the PCB and the NPCB. For the PCB, 90% of the measured points in the mesh have a distance to the CAD shorter than 0.16 mm. For the NPCB, 90% of the measured points of the NPCB have a distance to the CAD shorter than 0.11 mm, showing a 31% decrease of the C2M distance. The surface deviation is therefore smaller with the NPCB than with the PCB. Figure 5d shows a SEM picture of the surface of the PCB taken at a location close to the edge of the specimen, as indicated by the red dot on the schematic. The stacking of layers is visible, with the superposition of the 0° and 90° layers. However, the 90° layers exhibit a significant amount of over extrusion, as highlighted in the yellow rectangle. Figure 5e shows an enlargement of the NPCB upper facesheet located close to the edge. The waves observed on the scan appear to be created by the inappropriate overlapping of the 90° filaments, as shown in the yellow rectangle, which is the same issue encountered with the PCB. This over extrusion might be due to printing distance: at such a short printing distance, the 6-DOF robot could not reach its printing speed

but kept extruding as if it was printing at 15 mm/s. A more accurate synchronization of the extrusion flowrate with the actual speed of the 6-DOF robot could improve this issue.

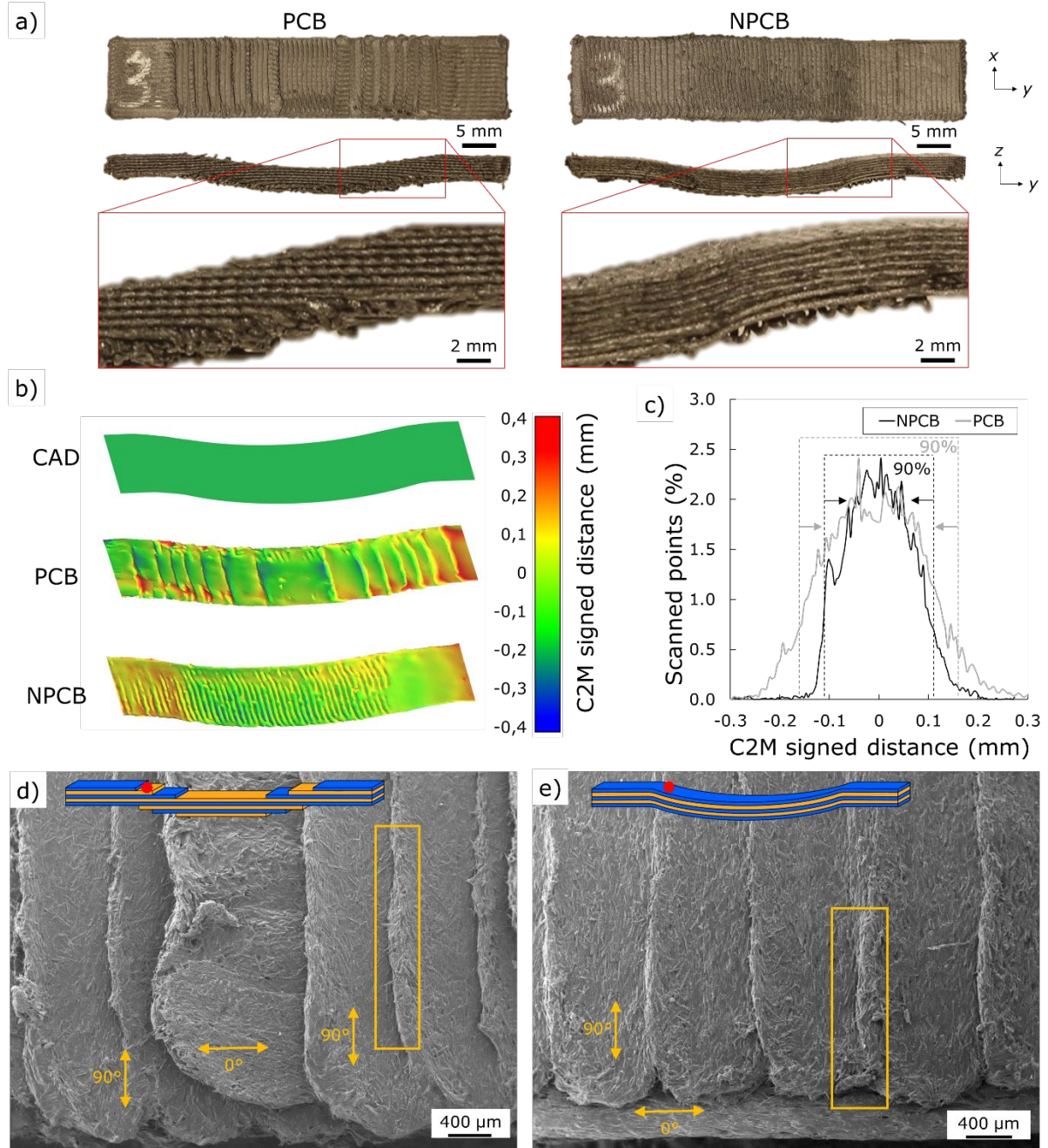


Figure 5: Geometrical comparison of the PCB and the NPCB a) Photograph of a PCB (left) and a NPCB (right) from top and side views, in which the layers are visible. The beams were additively manufactured using FFF with CFR-PEEK inside the heating enclosure. b) Comparison of the scanned surface of a PCB and a NPCB to their respective CAD model. The staircase effect is visible on the scan of the PCB, while the color gradient of the NPCB shows its surface is smoother. c) Comparison of the C2M signed distances of the PCB and the NPCB. 90% of the measured C2M distances of the PCB are lower than 0.16 mm, falling to 0.11 mm with the NPCB, confirming that the NPCB specimen has a lower surface deviation than the PCB. d) SEM picture of the surface of the PCB: the stacking of the 0° and 90° layers is visible, as well as some over extrusion in the 90° direction, due to a short distance extrusion. e) SEM

picture of the surface of the NPCB: due to over extrusion, the layers overlap on each other, creating the wavy surface visible on b) because of a short distance extrusion.

3.2. Analysis of the curved beams under 3-point bending testing

Figure 6a shows the 3-point bending test setup used. Figure 6b shows the load/displacement curves of the 5 tested specimens of PCB. The load values all linearly increased until failing between ~ 2.8 mm and ~ 3.4 mm of mid-span displacement for maximum loads ranging from ~ 190 N to ~ 220 N. Figure 6b also shows the load/displacement curves of the 5 tested specimens of NPCB. They all failed between ~ 2.6 mm and ~ 3.4 mm for maximum loads ranging from ~ 250 N to ~ 293 N, which are higher loads than for the PCB. The NPCB also exhibited a tougher and more gradual failure than the PCB. Almost all PCB loading curves immediately dropped at the first crack, showing a brittle failure, as opposed to NPCB specimens which did not completely break after the first load drop. None of the five NPCB tested specimens exhibited exterior signs of failure, unlike the PCB specimens, which either broke totally or exhibited cracks on their outer surface. Figure 6c compares the stiffness and the highest supported load of the PCB and the NPCB. The PCB and the NPCB respectively exhibit an average stiffness of 81.4 N/mm (95% confidence interval $CI_{95} = 2.1$ N/mm) and 105.1 N/mm ($CI_{95} = 2.3$ N/mm), which represents a 28 % higher value for the NPCB when compared to the PCB. The PCB and the NPCB respectively exhibit a highest supported load of 205.4 N ($CI_{95} = 24.9$ N/mm) and 264.2 N ($CI_{95} = 19.7$ N/mm), which represents a 29% higher value for the NPCB when compared to the PCB. Further investigations were made to analyze the failure of the beams. Figure 6d shows a SEM picture of the broken interface. It seems that the crack did not propagate vertically through the specimen but rather followed a staircase pattern. Most of the broken layers visible from this perspective were oriented along a 0° angle deposition, as confirmed by a zoomed SEM picture on Figure 6di. The broken fibers are visibly oriented along the yellow arrow, and so are the filaments. One 90° layer is visible on the right of the image, as shown by the red arrow. The orientation of the layer was confirmed by the SEM picture of Figure 6dii.,

in which the direction of the fiber generally follows the yellow arrow. This layer is visible because the 0° layer above delaminated. Figure 6e shows a CT-scan picture of a small portion of the fractured beam highlighted by the yellow square on the schematic of another tested PCB specimen. A crack running through the specimen in the yz plane is clearly visible. We can observe that the fracture propagated following the same kind of staircase pattern as the specimen of Figure 6c. The crack propagated vertically between a 90° layer and a 0° layer, then followed the interlayer interface in the direction of the applied load over a distance of ~ 650 μm , until it propagated again through the thickness in a 90° layer and a 0° layer, as shown by the red arrows. Figure 6f shows a CT-scan picture of a portion of a tested NPCB specimen highlighted by the yellow square on the schematic. No external sign of fracture could be found on any tested specimen, but delamination occurred inside, leading to the failure of the specimen. According to the scan, it seems that 0° layers delaminated from 90° layers below. These pictures provide insights about the different behaviors of the two types of beams. Both the PCB and the NPCB delaminated between a 90° layer and the 0° layer above. This could be due to the required printing time per type of layer: 90° layers were composed of many short lines of length ≤ 3 cm, preventing the 6-DOF robot from reaching its full targeted speed of 15 mm/s, while 0° layers were composed of longer lines and were then printed at full speed. When printing a 90° layer, the nozzle was slower and spent more time heating the deposition area, and thus probably increased the adhesion between each layer. A downside to this slower deposition was that once printed, the 90° layer had more time to cool down until the printing of the next 0° layer above. The 0° layer was printed faster and on a less heated surface, thereby increasing the risk of delamination between this layer and the previous 90° layer below. However, the NPCB specimen only delaminated between the layers near the mid-plane of the beam, while the PCB specimens failed due to crack propagation through the layers. An increase in the chamber

temperature to help reduce the cooling speed and enhance the adhesion is an interesting possibility to investigate for future work.

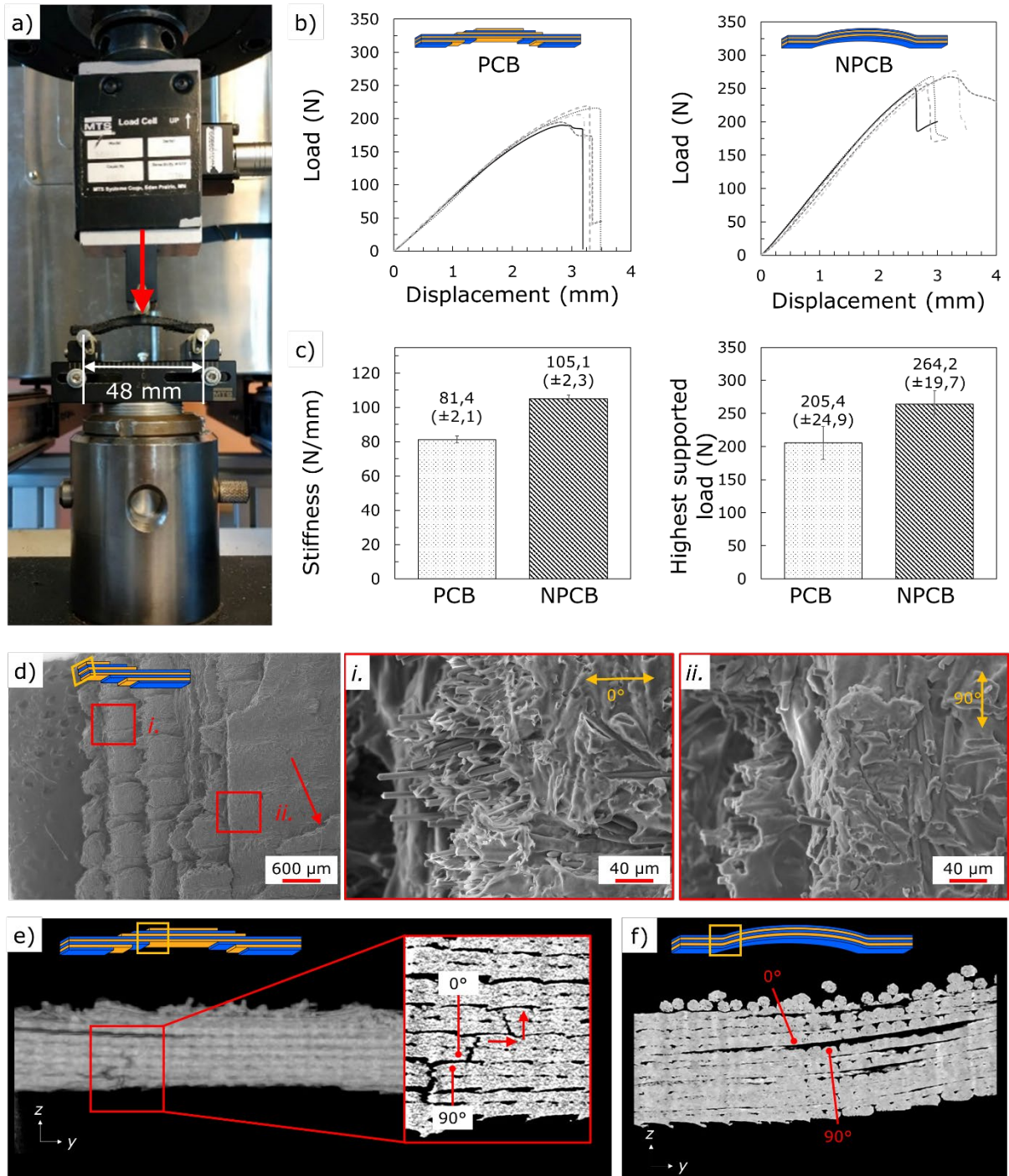


Figure 6: Comparison of the mechanical performance of the PCB and the NPCB in 3-point bending. a) Photograph of the 3-point bending test setup with a beam placed on the supports. b) Graph of the load/displacement of the 5 specimens of PCB (brittle and complete failure) and of the 5 specimens of NPCB (tougher and more gradual failure). c) Graph comparing the stiffness and the highest supported load of NPCB and PCB, with 95% confidence intervals indicated between parentheses: NPCB exhibits a stiffness 28% higher and a supported load 29% higher than PCB. d) SEM picture of the fractured surface of one PCB i) SEM picture of a 0° layer, ii) SEM picture of a 90° layer. e) CT-scan picture of a

PCB specimen: propagation of the fracture and delamination shown by the arrows on the inset f). CT-scan picture of a NPCB specimen: delamination area between a 90° layer and a 0° layer in the specimen.

3.3. Characterization of the non-planar curved sandwich panel

Non-planar curved sandwich panels have been manufactured inside the heating enclosure using the CFR-PEEK material. Figure 7a shows the printing time distribution and the number of layers in each type of zone in the panel. Although there are only 20 layers of the walled scaffold (12% of the total numbers of layers), this zone took a total of 300 minutes to print (36% of the printing time). The stop-and-go due to the walled scaffold prevented the 6-DOF robot from reaching its full speed. The lower/adaptative facesheet and the microscaffold 30%, however, represent respectively 16% and 23% of the total number of layers, and just needed respectively 18% and 24% of the printing time (i.e., 150 minutes and 200 minutes) to be manufactured. The core also had numerous stop-and-gos due to its hollow geometry: it represents almost 47% of the number of layers (80 layers) but could be printed in 180 minutes (20% of the printing time). A way to improve the manufacture of the panel would be to optimize the toolpath of the 6-DOF robot to reduce the stop-and-go by rounding the corners of the trapezoidal structures.

Figure 7b shows a portion of a modified non-planar sandwich panel which was printed to display the features inside the panel. In Figure 7bi., the partitioned microscaffold whose infill target is 74% is partially clogged. An example of clogged pores is highlighted in the yellow square, while unclogged pores are visible in the green square. The unclogged pores in the picture have an average surface of $\bar{A} = 0.03 \text{ mm}^2$ ($CI_{95} = 0.02 \text{ mm}^2$), for an average filament width of $\bar{p} = 393 \text{ } \mu\text{m}$ ($CI_{95} = 46 \text{ } \mu\text{m}$), representing an infill percentage of 70%. In the lower part of the picture, the filaments are fused together, and their average width is $\bar{p} = 414 \text{ } \mu\text{m}$ ($CI_{95} = 11 \text{ } \mu\text{m}$). The printability index S of the scaffold is 0.78, which is not considered as an acceptable value because too many defects could affect the acoustic performance and weaken the structure. On Figure 7bii., the microscaffold 30% has an average filament width of $\bar{p} =$

401 μm ($\text{CI}_{95} = 27 \mu\text{m}$) and an average pore surface $\bar{A} = 0.80 \text{ mm}^2$ ($\text{CI}_{95} = 0.07 \text{ mm}^2$), which means an infill percentage of 31%, very close to the targeted 30%. The printability index S is 0.97, which is higher than the fixed threshold of 0.95. The difference between the two scaffolds could be explained by two aspects: the walled micro scaffold 74% is denser, and partitioned. The walled partitioning of the micro scaffold 74% reduced the printing distance, thereby inducing an over extrusion and then blocking the pores, which is not the case for the micro scaffold 30% that is not walled. A similar phenomenon happened with the curved beams detailed in Section 3.1. Figure 7biii. shows an optical microscope picture of the flat sidewall of a panel, at the level of the trapezoidal core. The average layer thickness is $\bar{p} = 200 \mu\text{m}$ ($\text{CI}_{95} = 12 \mu\text{m}$) and no major defects are visible. Figure 7c shows a picture of a 3D printed CFR-PEEK sandwich panel, with a zoomed in picture showing the side of the panel. Although the vertical walls above the trapezoidal structures can be considered well printed, the inclined walls of the acoustic core present more defects (red and blue arrows). The red arrows show holes on the edge of a trapezoidal cell. The filaments did not reach the end of their extrusion path. This could be due to the deposition speed during the back-and-forth printing pattern being too high. Avoiding this toolpath would help solve this issue without reducing the speed. Rounding the sharp corners of the trapezoidal shapes should also help improving the deposition of the material and reducing the defects highlighted by the red arrows. The blue arrows show entire layers that did not stick to the underlying layers on an inclined wall of a trapezoidal cell, thereby creating a gap between layers. The 6-DOF robot seemed to struggle extruding a single filament along an inclined direction. However, along the z axis, it could perfectly print a single extrusion wall, as seen in Figure 7biii. To keep such an inclined wall, a solution would be to decrease the printing speed when manufacturing these specific structures to reduce the production of defects, or to build a larger wall consisting of at least two filaments of thickness.

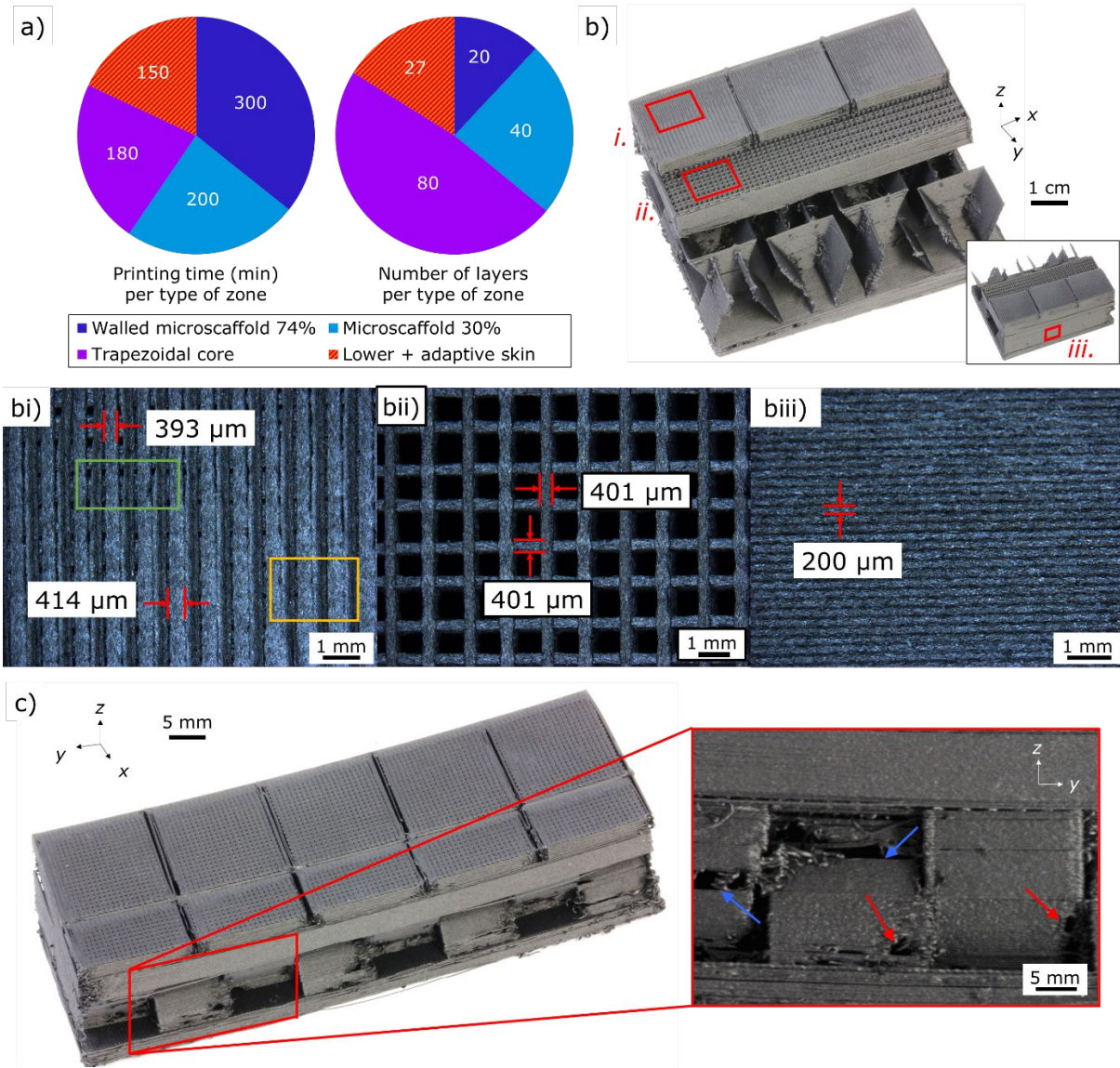


Figure 7: Geometry of the non-planar sandwich panel. a) Pie charts of the printing time distribution and the number of layers for each zone of the non-planar panel. b) Photograph of a piece of non-planar curved sandwich panel printed to display the internal layers: i) Optical microscope picture of the upper facesheet of the panel (walled microscaffold 74%), in which many pores are clogged due to a mismatch between the path distance and the material flowrate, with a scaffold printability index $S = 0.78$. ii) Optical microscope picture of microscaffold 30%, in which filaments have an average diameter of 401 (± 14) μm and a scaffold printability index $S = 0.97$, iii) Optical microscope picture of the side of the specimen, in which the layers have a constant thickness of 200 μm . c) Photograph of the CFR-PEEK non-planar sandwich panel inclined walls. Two types of defects: printing path (red), angled superposition of single extrusion filaments (blue). However, the vertical wall above presents no visible defects.

Figure 8a shows a comparison of the CAD of the sandwich panel to its scanned surface on the printed demonstrator, on which a color scale indicates the C2M signed distances. On the top surface, the longest measured C2M signed distances are 0.5 mm and -0.3 mm. These maximum values can be found in the upper corners of the panel, as shown in green in Figure 8a. The

overall maximum C2M signed distance of -2.5 mm was found on one of the short sides of the panel (in blue on the figure). Figure 8b shows a graph of the distribution of the C2M signed distances of the upper skin of the panel: the peak is narrow and centered in 0. Indeed, ~90% of the distances were comprised between ± 0.22 mm, which is acceptable considering the overall dimension of the panel and the fact that a single layer has a thickness of 0.2 mm. Figure 8c shows a graph of the distribution of the C2M signed distances of the curved sides: ~90% of the distances are comprised between ± 0.55 mm. The distance is higher than with the upper skin but is similar to what was found on a planar print of acoustic cells in Supplementary Figure S3. The surface deviation of the curved side is acceptable, although it could probably be lowered with a more accurate calibration of the 6-DOF robot. Figure 8d shows the graph of the distribution of the C2M signed distances of the short sides of the panel. This graph does not have a narrow peak centered in 0 like the two previous graphs do, but rather spreads between -1.5 mm and 1 mm with peaks around -1.5 mm, -1 mm, 0 mm and 1 mm. When looking at Figure 8a, the C2M distances of the short sides are not uniform: some large areas are green, yellow or blue and they are each time delimited by their intersection with the core (the pattern of the trapezoidal shape is visible on the scan, which is not the case with the CAD part). Each type of color would represent a peak on the graph. Since each area has a uniform color, the difference of C2M distance was attributed to a misinterpretation of the custom slicer rather than to warping or deformation occurring during the print. The short side was not interpreted as a uniform flat surface but rather as a combination of smaller surfaces separated by the intersections with the vertical walls of the trapezoidal shapes, inducing a shift between their position in the CAD file and their position once printed.

The complex printed structure (e.g., with many overhangs, single extrusion walls, non-planar deposition, variation of extrusion thickness through the same layers), composed of several types of non-planar stackings (e.g., plain layers, trapezoidal core, microsccaffold), fitted the double

curvature surface of the aircraft casing. A larger nozzle could reduce the printing time, however the thickness of the walls in the core would increase, and the filaments in the microscaffold would be thicker. A compromise must be found between the minimum extrusion feature and the manufacturing time. Two printheads (one large and one small) could be implemented, but a system to calculate and avoid the collisions due to the change of toolhead and non-planar specimens would be required. The tool could also be changed during the manufacture, using first a large nozzle for the adaptive layer and then a thin nozzle for the acoustic. Using a non-planar heating bed could reduce the printing time as printing supports would no longer be required. However, it means a custom bed would be required for each different non-planar print.

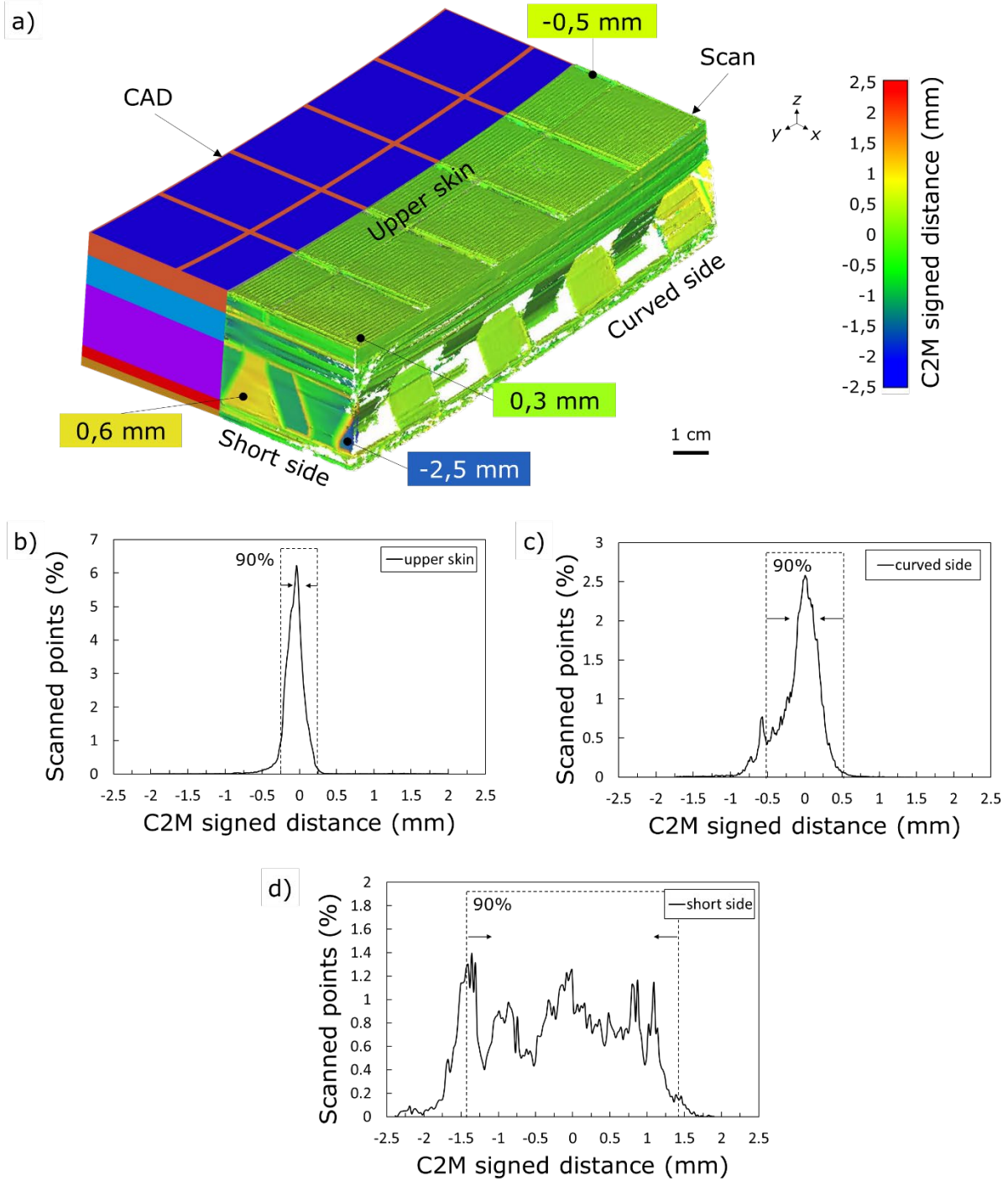


Figure 8: Dimensional analysis of the non-planar curved sandwich panel a) Comparison of the reference CAD of a non-planar sandwich panel with its scanned surface. The highest C2M signed distance on the upper skin is ± 0.28 mm. b) Graph of the C2M signed distance of the upper surface of the non-planar sandwich panel: 90% of the measured distances are comprised between ± 0.22 mm. c) Graph of the C2M signed distance of the upper surface with the acoustic cells of the non-planar sandwich panel: 90% of the measured distances are comprised between ± 0.55 mm. d) Graph of the C2M signed distance of the upper surface of the non-planar sandwich panel: 90% of the measured distances are comprised between ± 1.45 mm, but this larger variation is due to the slicing method.

4. Conclusion

A 6-DOF robotic platform with a heating enclosure was successfully used to enable the non-planar additive manufacturing of HTRT structures. NPCB specimens have shown a ~31% decrease in the surface deviation compared to PCB specimens, as well as a ~28% increase in stiffness during a 3-point bending test. A sandwich structure composed of 6 distinct zones was also manufactured using the 6-DOF robot and the enclosure, thereby showing the versatility of the 3D printing infrastructure and some of its current limitations. The robotic system was capable of printing sophisticated non-planar structures with high precision, such as single wall extrusions and microscaffolds inside the same structure, although short distance extrusion (e.g., corners of the trapezoidal core) and inclined single extrusion (e.g., sidewalls of the core) are still a challenge. Future work will focus on adjusting the extrusion to the 6-DOF robot's real speed during small extrusion distances to reduce over extrusion and improve the printability of dense scaffolds. A higher and more homogenous temperature is also a crucial point to improve. Another task will be to increase the printing speed by investigating pellets extruders, and therefore legitimizing the use of a robotic platform for large scale non-planar additive manufacturing. This research contributes to push the limits of HTRT non-planar FFF by presenting a process to manufacture multi-functional sandwich panels. The parts printed with our setup could have direct applications for custom noise-reduction production in the aerospace field. Non-planar HTRT scaffold printing could also have applications in the biomedical field.

Acknowledgments

The authors acknowledge the financial support from the FACMO Chair supported by the Natural Sciences and Engineering Research Council of Canada (NSERC, CRDPJ 514761-1), the Canada Foundation for Innovation (CFI, Project #36450), the Government of Québec (Project #36450), and Safran Group (France). The authors also acknowledge the help of Marie-Hélène Bernier with the CT-scan measurements, Claire Circlé and Dogan Aslan with the SEM

measurements, Yanik Landry-Ducharme with technical assistance, and Arthur Sadaa for the training on DragonFly.

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Supplementary Material

Enclosure temperature measurements

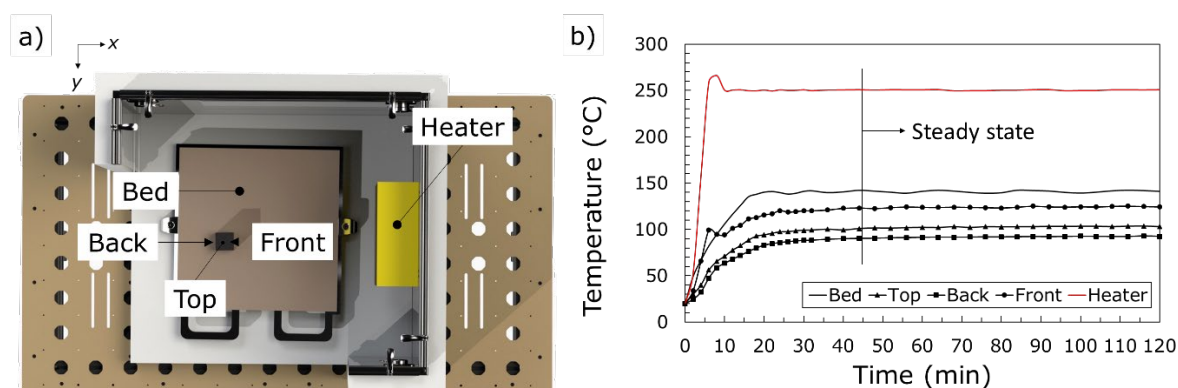


Figure S1: Temperature measurements inside the heating enclosure. a) Top view schematic of the enclosure with the locations of the different thermocouples. Three thermocouples are fixed on a cubic CFR-PEEK printed part: one on the face facing the heater (Front), one on its opposite side (Back), and one on the top surface (Top). Another thermocouple is taped to the bed, close to the cube (Bed), and another one measures the temperature of the heater. b) Graph of the temperature measured by five different thermocouples inside the enclosure. The temperature of each thermocouple stabilizes after 45 minutes of heating.

Table S2 Average temperatures measured inside the heating enclosure after temperature stabilization (45 minutes), with the standard deviation indicated between parentheses.

Thermocouple name	Average temperature between 45 and 120 min (°C)
Front	124.0 (± 0.7)
Back	91.7 (± 0.8)
Top	102.4 (± 1.0)
Bed	140.7 (± 1.1)

Core of the sandwich panel: acoustic shapes

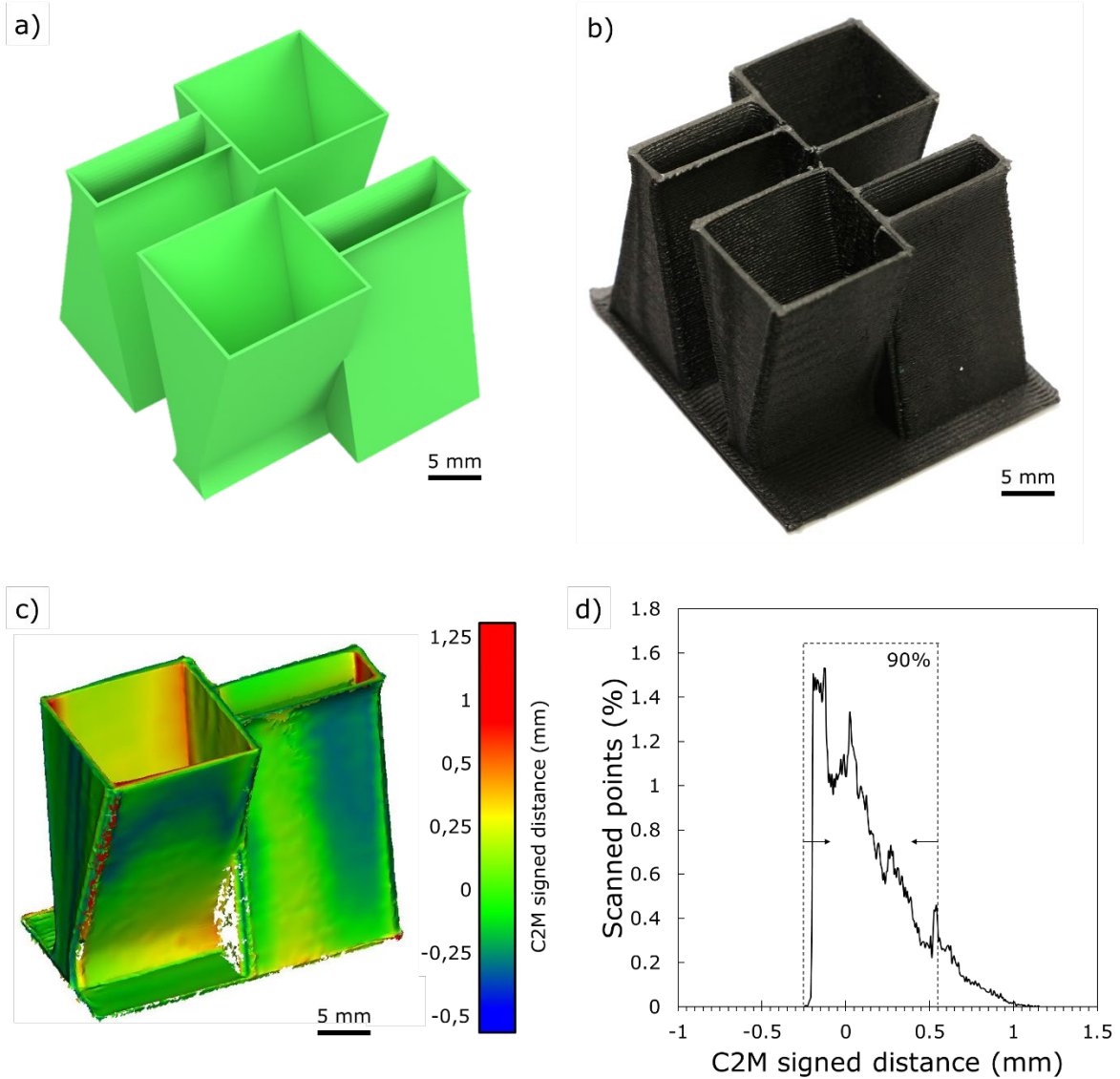


Figure S3: Hollow trapezoidal shapes a) Schematic of the acoustic cell composed of four trapezoidal hollow shapes alternatively turned upside down. b) Picture of a cell printed with CFR-PEEK inside the heating enclosure. c). Comparison of the scanned surface of a 3D-printed half cell to its CAD model. Over extrusion is visible in light red in the internal edges of the trapezoidal hollow shape. d) Graph of the C2M signed distance of the half-cell: 90% of the measured distances are comprised between -0.25 mm and +0.55 mm.